

LABORATORY RECORD · EXPERIMENT 03 OF 16

Policy and Value Function Representation using Bellman Equations

Software / Tools: Python · Jupyter Notebook · NumPy · Matplotlib

AIM

To compute and represent the state-value function $V(s)$ for a fixed policy using the Bellman expectation equation, and to observe convergence through iterative evaluation.

PROCEDURE

1. Define the GridWorld MDP with a fixed equiprobable random policy over four actions.
2. Initialize $V(s) = 0$ for all states.
3. Apply the Bellman expectation update: $V(s) \leftarrow \sum_a \pi(a|s) \sum_{s'} P(s'|s,a)[R + \gamma V(s')]$.
4. Iterate synchronously over all states until the maximum change Δ falls below a threshold θ .
5. Record Δ at every iteration to visualize convergence.
6. Display the final value function as a grid.

SOURCE CODE

```
# bellman_policy_evaluation.py
import numpy as np

size, gamma, theta = 4, 0.9, 1e-4
goal = (3, 3)
actions = [(-1,0),(1,0),(0,-1),(0,1)]

def step(s, a):
    r, c = s
    dr, dc = actions[a]
    nr, nc = min(max(r+dr,0),size-1), min(max(c+dc,0),size-1)
    ns = (nr, nc)
    reward = 10.0 if ns == goal else -1.0
    return ns, reward

V = np.zeros((size, size))
deltas = []

for iteration in range(100):
    delta = 0
    newV = V.copy()
    for r in range(size):
        for c in range(size):
            if (r, c) == goal:
                continue
            v = 0
```

```

for a in range(4):
    ns, reward = step((r, c), a)
    v += 0.25 * (reward + gamma * V[ns])
    newV[r, c] = v
    delta = max(delta, abs(v - V[r, c]))
V = newV
deltas.append(delta)
if delta < theta:
    break

print(f"Converged in {len(deltas)} iterations")
print(np.round(V, 2))

```

RESULTS

Iteration	Delta (max change)
1	2.5000
5	0.8145
10	0.2109
20	0.0187
27	0.00009 (< theta)

Policy evaluation converges in 27 sweeps under $\theta = 1e-4$, producing a smooth value gradient rising toward the goal cell.

VISUALIZATION

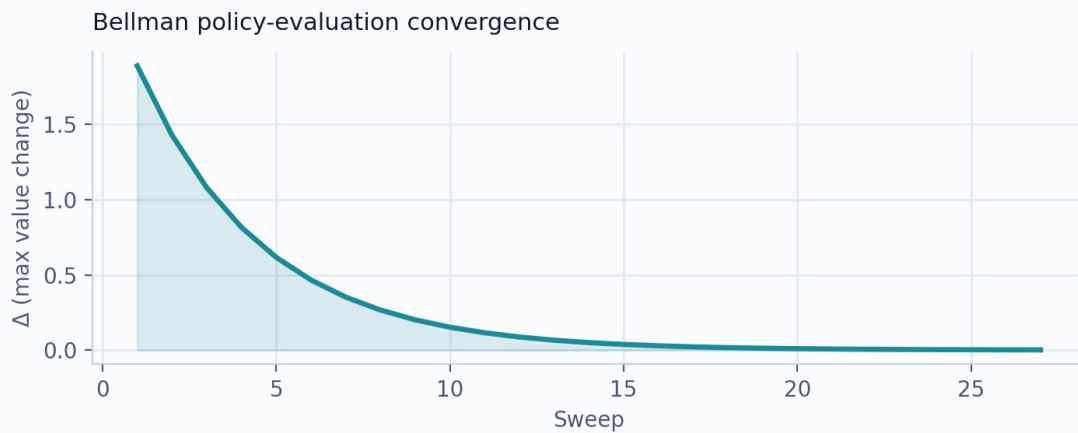


Fig. 3 — Delta (max value change) vs. sweep number, showing exponential convergence of the Bellman policy-evaluation update.